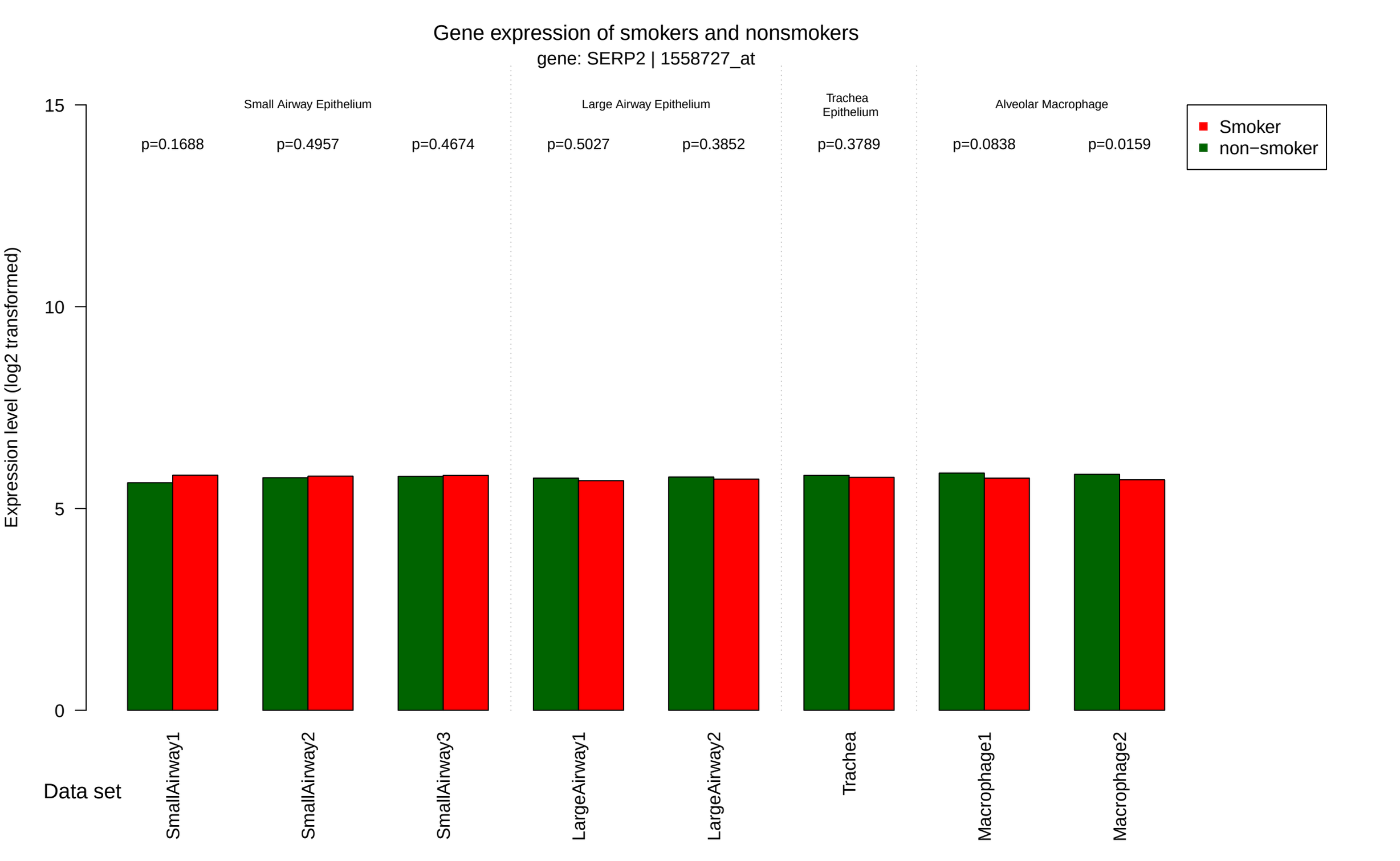
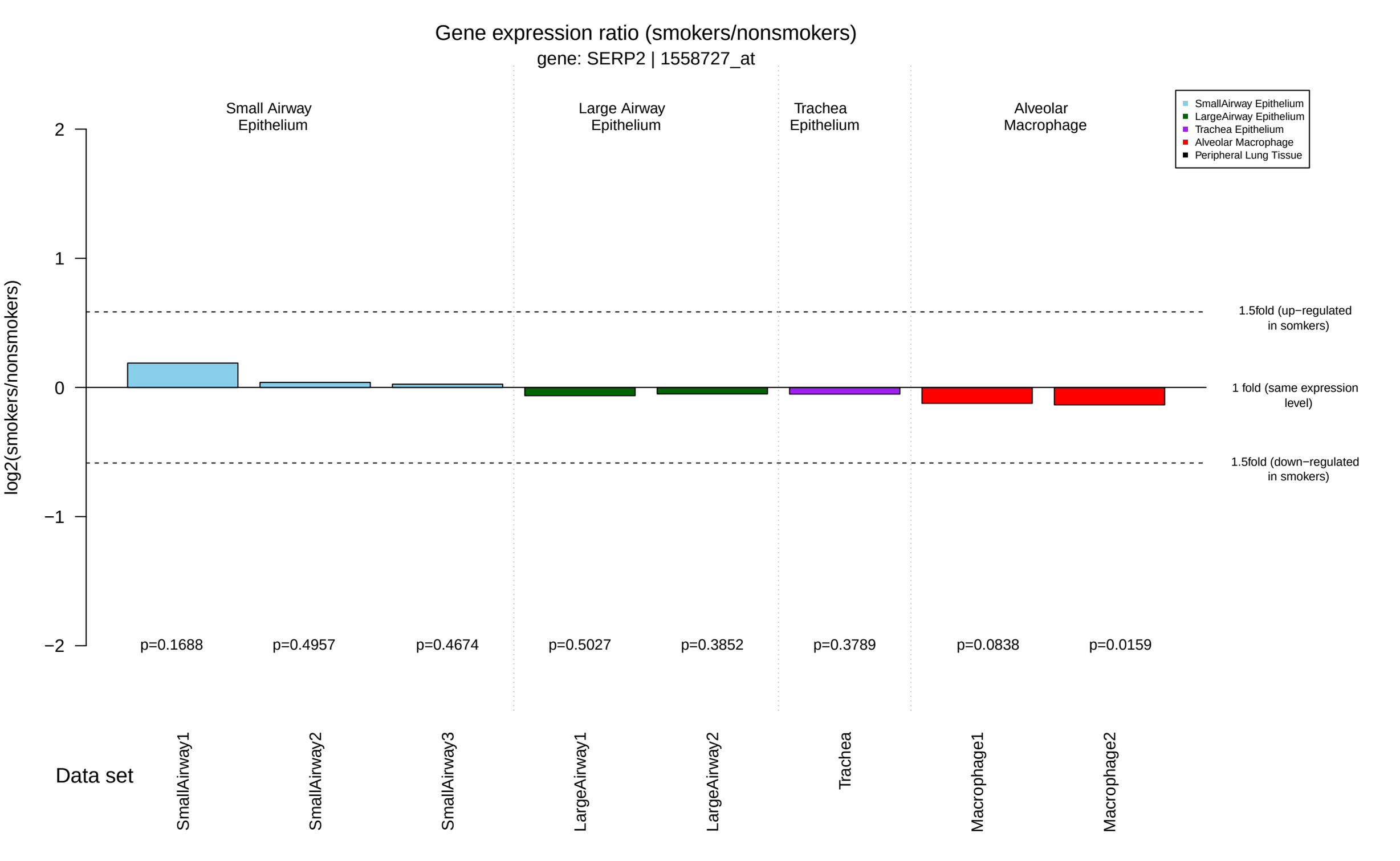
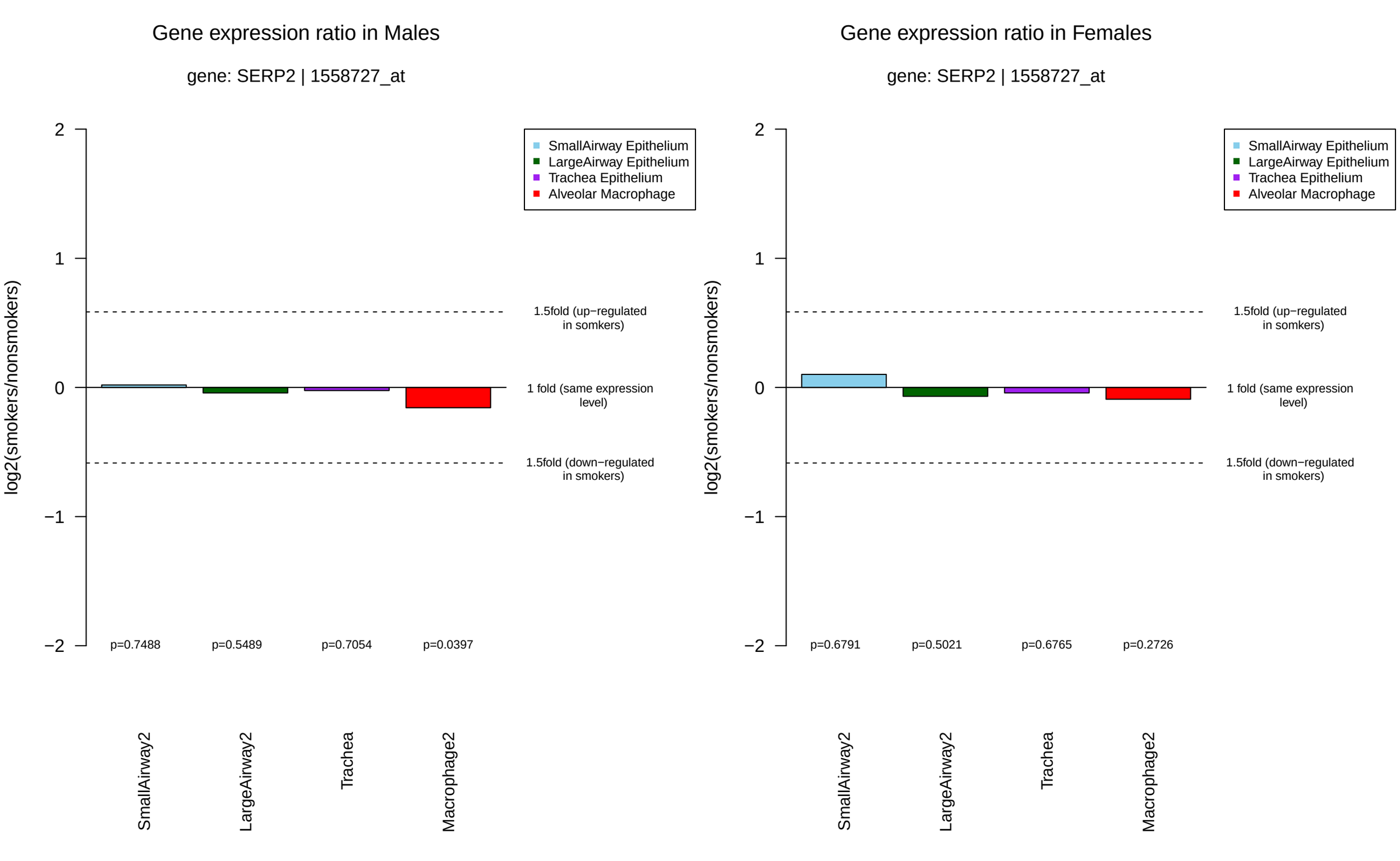
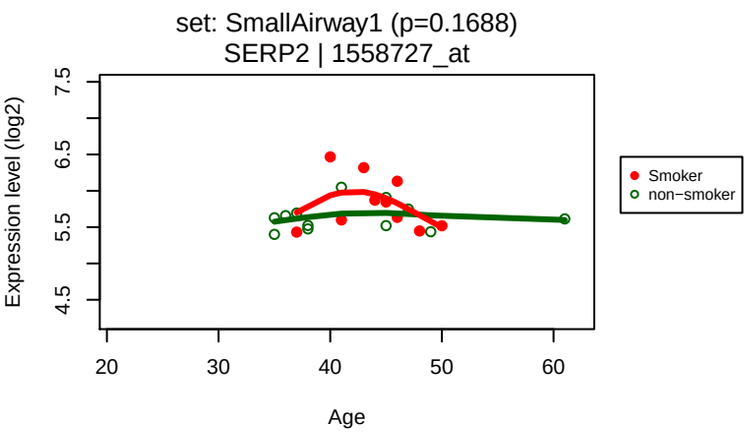
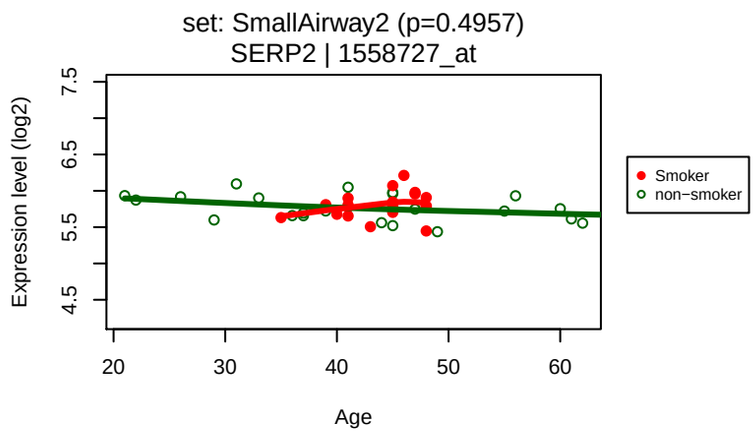
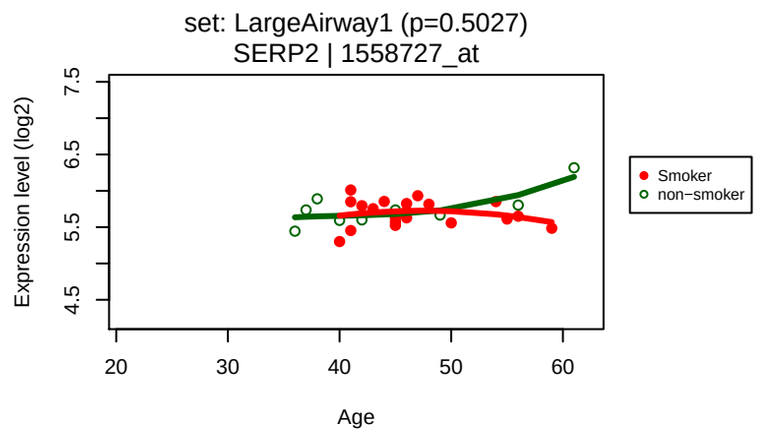

Gene expression of smokers and nonsmokers
gene: SERP2 | 1558727_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1688
p=0.4957
p=0.4674
p=0.5027
p=0.3852
p=0.3789
p=0.0838
p=0.0159
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

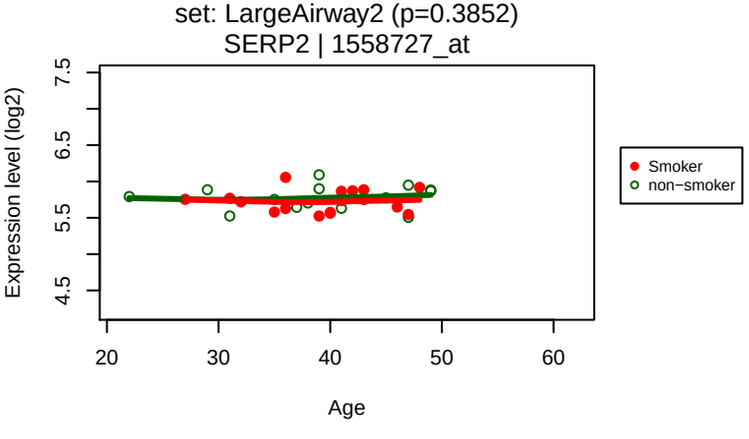

Gene expression ratio (smokers/nonsmokers)
gene: SERP2 | 1558727_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1688
p=0.4957
p=0.4674
p=0.5027
p=0.3852
p=0.3789
p=0.0838
p=0.0159
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

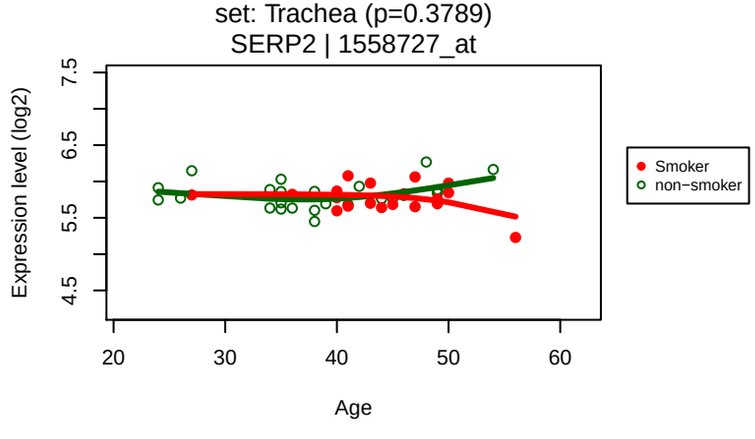

Gene expression ratio in Males
gene: SERP2 | 1558727_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7488
p=0.5489
p=0.7054
p=0.0397
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SERP2 | 1558727_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6791
p=0.5021
p=0.6765
p=0.2726
SmallAirway2
LargeAirway2
Trachea
Macrophage2

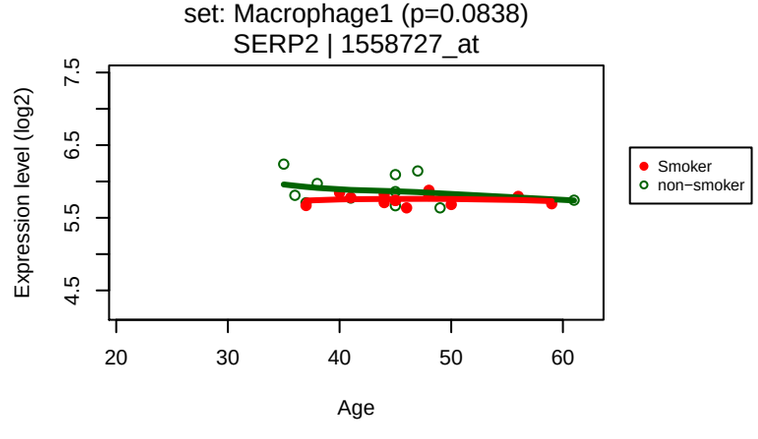

set: SmallAirway1 (p=0.1688)
SERP2 | 1558727_at
Expression level (log2)
4.5
5.5
6.5
7.5
20
30
40
50
60
Age
Smoker
non−smoker

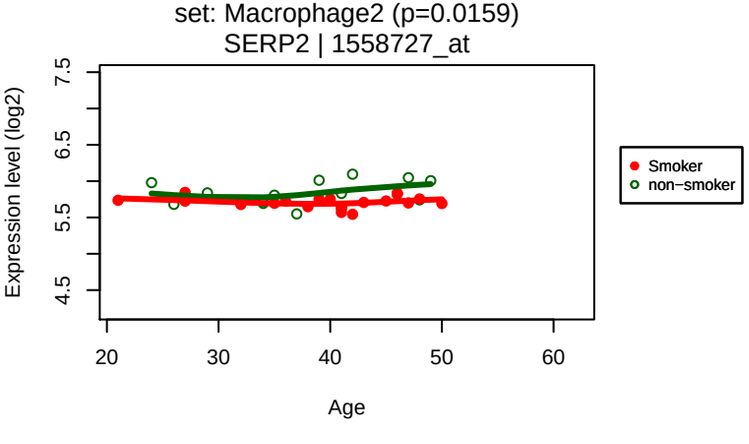

set: SmallAirway2 (p=0.4957)
SERP2 | 1558727_at
Expression level (log2)
4.5
5.5
6.5
7.5
20
30
40
50
60
Age
Smoker
non−smoker

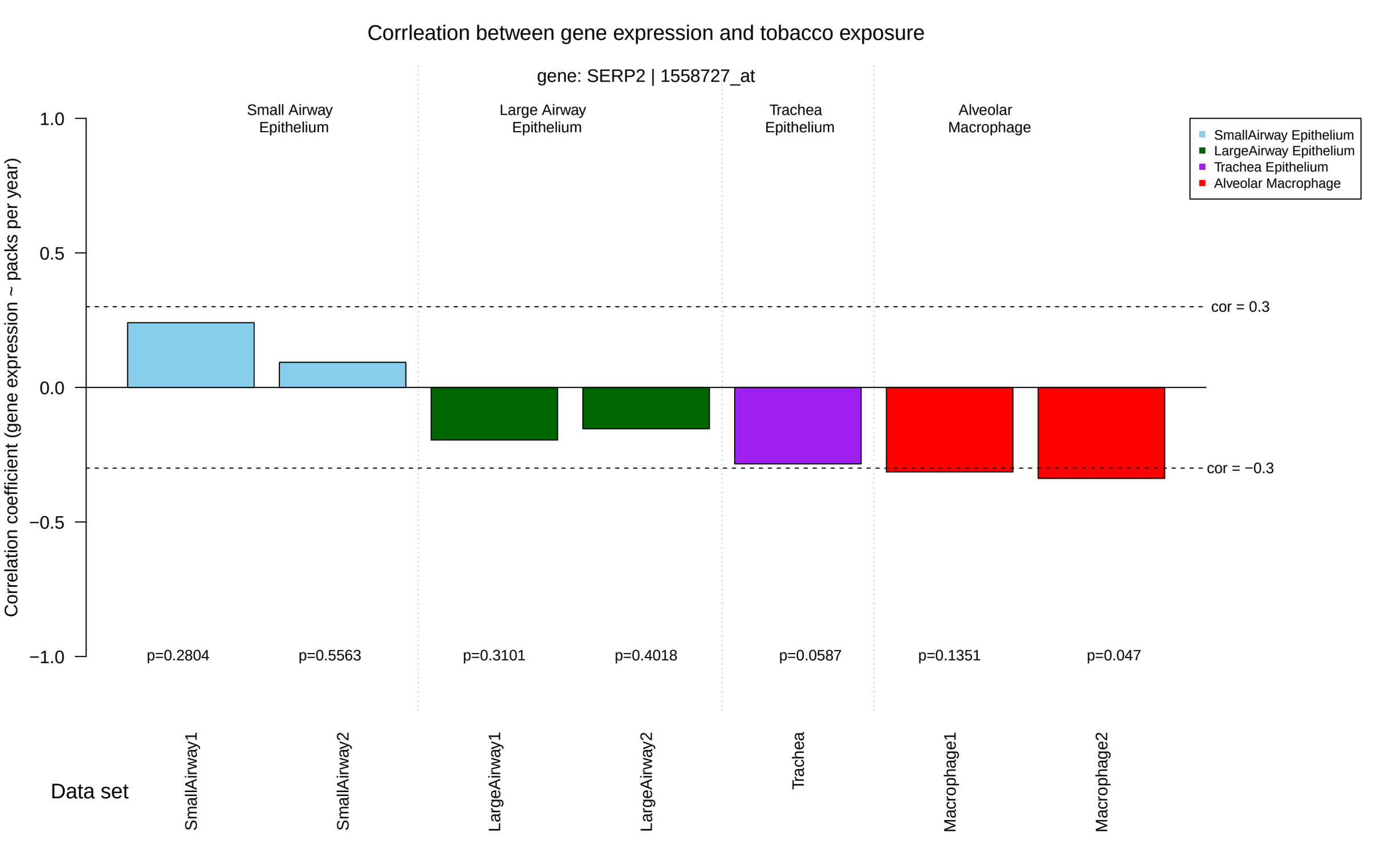

set: LargeAirway1 (p=0.5027)
SERP2 | 1558727_at
Expression level (log2)
4.5
5.5
6.5
7.5
20
30
40
50
60
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3852)
SERP2 | 1558727_at
Expression level (log2)
4.5
5.5
6.5
7.5
20
30
40
50
60
Age
Smoker
non−smoker


set: Trachea (p=0.3789)
SERP2 | 1558727_at
Expression level (log2)
4.5
5.5
6.5
7.5
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0838)
SERP2 | 1558727_at
Expression level (log2)
4.5
5.5
6.5
7.5
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0159)
SERP2 | 1558727_at
Expression level (log2)
4.5
5.5
6.5
7.5
20
30
40
50
60
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SERP2 | 1558727_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2804
p=0.5563
p=0.3101
p=0.4018
p=0.0587
p=0.1351
p=0.047
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2